James Cook University
Human Research Ethics Committee

HUMAN ETHICS APPROVAL NUMBER	H
Version No. and Date*	

HUMAN RESEARCH ETHICS APPLICATION

Submission and completion guide:

- ***Please complete the checklist, application, Ethics Advisor Review and supporting documents. Please then combine as 1 PDF by email to ethics@jcu.edu.au***
- ***You must not start your project until you receive formal approval from the HREC (that includes any advertising or recruitment)***
- ***Please read the instructions on how to complete a Human Ethics Advisor Review on the last page of this document***
- ***Note the HREC meeting submission dates to ensure you will meet the submission deadline.***

Ensure you addressed the feedback given by the Human Ethics Advisor and highlighted these changes in yellow? (See instructions on the last page)	☐Y	☐N

***Each time the application and/or supporting documents are updated please change the version number and date of each document.**

Before starting your application, please read the National Health and Medical Research Council, *National Statement on Ethical Conduct in Human Research, 2007 (Update 2018)*

A. HUMAN RESEARCH RISK ASSESSMENT CHECKLIST

This checklist is to help assess potential risks involved in conducting your planned human research project and also be used to decide which level of ethical review your application will require (Refer to the *National Statement Chapter 2.1*).

1. Go through the checklist and check 'YES' or 'NO' for each of the risks listed.
2. Based on this risk assessment, assign a risk category in *Question 7*.
3. For each 'YES' you have identified a potential risk, consider how you will:
   a. Gauge the risk – consider the likelihood and severity of any risk.
   b. Remove the risk – can you remove the risk by change your project's methods?
   c. Minimise the risk – If the risk can't be removed, can it be minimised by a change to the methods?
   d. Manage the risk/harm – can any remaining risk or harm be managed by putting in place a monitoring program to detect and institute measures to deal with any harm?
4. Outline in your application how you will minimise and manage the risks identified.

A. Contentious or Controversial Topics: Are any of the following topics covered in part or in whole in your project?				
Research about parenting issues	☐	YES	☒	NO
Research investigating sensitive personal issues	☐	YES	☒	NO
Research investigating sensitive cultural issues	☐	YES	☒	NO
Explorations of grief, death or serious/traumatic loss	☐	YES	☒	NO
Depression, mood states, anxiety	☐	YES	☒	NO
Gambling	☐	YES	☒	NO
Eating disorders	☐	YES	☒	NO
Illicit drug use	☐	YES	☒	NO
Substance abuse	☐	YES	☒	NO
Self-reporting of criminal behaviour	☐	YES	☒	NO

A psychological disorder	☐	YES	☒	NO
Suicide	☐	YES	☒	NO
Gender identity	☐	YES	☒	NO
Sexuality	☐	YES	☒	NO
Race or ethnic identity	☐	YES	☒	NO
Any disease or health problem	☐	YES	☒	NO
Fertility	☐	YES	☒	NO
Termination of pregnancy	☐	YES	☒	NO

B. Research procedures/practices: Are any of the following procedures/practices to be used in your project?				
Use of students' personal information from a JCU database without specific consent from the students. *If 'YES' please submit your application to the JCU Privacy Officer to approve before the HREC*	☐	YES	☒	NO
Use of personal data obtained from Commonwealth or State Government Department/Agency	☐	YES	☒	NO
Use of personal data obtained from State Government Department/Agency	☐	YES	☒	NO
Use of personal information from a non-government organisation	☐	YES	☒	NO
Does your research require a waiver of consent?	☐	YES	☒	NO
Deception of participants	☐	YES	☒	NO
Concealing the purposes of the research	☐	YES	☒	NO
Use of human biospecimens in laboratory-based research	☐	YES	☒	NO
Genomic research	☐	YES	☒	NO
Covert observation	☐	YES	☒	NO
Audio or visual recording without consent	☐	YES	☒	NO
Recruitment of a third party or agency	☐	YES	☒	NO
Withholding from one group specific treatments or methods of learning, from which they may "benefit" (e.g. in medicine or teaching)	☐	YES	☒	NO
Psychological interventions or treatments	☐	YES	☒	NO
Administration of physical stimulation	☐	YES	☒	NO
Physically invasive procedures	☐	YES	☒	NO
Potentially painful procedures	☐	YES	☒	NO
Administration of drugs	☐	YES	☒	NO
Administration of other substances	☐	YES	☒	NO
Exposure to ionising radiation	☐	YES	☒	NO
Tissue sampling or blood taking	☐	YES	☒	NO
Collecting body fluid	☐	YES	☒	NO
Use of medical records where participants can be identified or linked	☐	YES	☒	NO
Genetic testing/DNA Extraction	☐	YES	☒	NO
A clinical trial or drug trial	☐	YES	☒	NO

C. Participant groups requiring special consideration: Does your project specifically target participants from any of the following groups?				
Patients, staff, data, facilities or sites in a Queensland health facility. *Please submit your application to the HREC of that facility*	☐	YES	☒	NO
Aboriginal and Torres Strait Islander Peoples *Please complete the Aboriginal and Torres Strait Islander peoples research supplement and attach it to this application*	☐	YES	☒	NO
People suffering from a psychological disorder	☐	YES	☒	NO
People suffering a physical vulnerability	☐	YES	☒	NO
People highly dependent on medical care who may be unable to give consent	☐	YES	☒	NO
Children or young people	☐	YES	☒	NO
People whose ability to give consent is impaired	☐	YES	☒	NO
Resident of a custodial institution	☐	YES	☒	NO

People unable to give free informed consent because of difficulties in understanding information provided eg. Language difficulties	☐	YES	☒	NO
People with a cognitive impairment, an intellectual disability or mental illness	☐	YES	☒	NO
People who may be involved in illegal activities	☐	YES	☒	NO
People in dependent or unequal relationships eg. Lecturer/student, doctor/patient, teacher/pupil & professional/client	☐	YES	☒	NO
Members of a socially and/or culturally identifiable group with special social/cultural/ethnic or religious beliefs or political vulnerabilities	☐	YES	☒	NO
Women who are pregnant and the human foetus	☐	YES	☒	NO
Participants are identifiable in the final report when specific consent for release has not been given	☐	YES	☒	NO

D. Risks to researchers: Are there any of the following risks to researchers involved?				
Are there any potential risks to the researcher? (e.g. research conducted in unsafe environments or trouble spots)?	☐	YES	☒	NO
Are there any potential risks to non-participants in the research, such as, participant's family members and social community? e.g. effects of biography on family and friends or infectious disease risk to the community)	☐	YES	☒	NO

E. Research overseas: Does your project involve researching in an overseas country?				
Where research is being undertaken in a politically unstable area	☐	YES	☒	NO
Where research involves sensitive cultural/social/political/ethnic/economic or religious issues	☐	YES	☒	NO
Where criticism of the government and institutions may be a risk to participants and/or researchers	☐	YES	☒	NO

F. Researcher conflict of interest: Do any of the following apply to your research?				
Are any of the researchers affiliated with any of the external organisations involved in your research?	☐	YES	☒	NO
Are any of the researchers in receipt of any financial benefit from any of the external organisations involved in your research?	☐	YES	☒	NO

G. Risk category: Assess your project's overall risk category (see the HREC's Application Guide for more information and examples)*	
☒ **1**	**Negligible risk:** Research in which there is no foreseeable risk of harm or discomfort and any foreseeable risk is no more than inconvenience.
☐ **2**	**Low risk:** Research in which the only foreseeable risk is one of discomfort. Discomforts include, for example, minor side-effects of medication, the discomfort of measuring blood pressure or the anxiety induced by an interview.
☐ **3**	**More than low risk:** Research with the potential to cause mild psychological distress or physical stress. Minor deviation from frank disclosure of the true nature of the research may be involved.
☐ **4**	**More than low risk:** Research with the potential to cause genuine but not severe psychological distress or physical pain with no long-term effects. Deception may be involved regarding the true nature of the research.
☐ **5**	**More than low risk:** Research with the potential to cause psychological distress or physical pain. Substantial deception may be involved.
☐ **6**	**More than low risk:** Research involving vulnerable participants; at risk populations; or research that may pose serious ethical considerations.

*Please note that if you have said 'Yes' to several items in the checklist it is likely that the research is more than low risk (H3-6). Assessing the risk of your research as being lower than it is may lead to a delay in the review of your application, as it will be sent for Executive Review, before being redirected to the HREC meeting.

HUMAN RESEARCH ETHICS APPLICATION FORM

1. **PROJECT TITLE**
   The project title should be in lay language and understandable by the participant groups being targeted.

Ironic effects of thought suppression – a multilab registered replication report

2. **PROJECT DURATION (in years)**
   If you are requesting more than 3 years, please justify why you will need more than 3 years and how you will have resources to cover this work.

Jan 2022 – Dec 2024

3. **RESEARCHER AND SUPERVISOR DETAILS, ROLE AND EXPERIENCE**

Principal Investigator	Deming (Adam) Wang		
Email:	adam.wang@jcu.edu.au		
Phone:	+65 82467624	**Mobile:**	+65 82467624
Discipline/College/Organisation	James Cook University – College of Healthcare Sciences		
Organisation Unit (ORGU) No.	2125		
Relationship to JCU (staff, student, non-JCU)	Staff		
If you are a student, provide the name of the degree being undertaken (and details of your supervisor below)			
Qualifications	PhD		
Role in the project What will be your role in the project?			
Lead investigator – this includes research conceptualisation, design, data collection, analyses, and article write up			
Experience Outline your experience in the role or outline how you will be trained and supervised			
I have published articles in peer-reviewed journals in the field of social psychology, and have experience running experimental research			

Co-investigator			
Email:			
Phone:		**Mobile:**	
Discipline/College/Organisation			
Organisation Unit (ORGU) No.			
Relationship to JCU (staff, student, non-JCU)			
If you are a student, provide the name of the degree and details of your supervisor below			
Qualifications			
Role in the project What will be your role in the project?			

Experience Outline your experience in the role or outline how you will be trained and supervised

Cut and paste extra tables required for additional researchers.
Remove un-used tables

Please outline below any additional personnel who will be assisting with research, their roles, experience and qualifications.

4. FUNDING AND FINANCIAL BENEFITS

4.1 Researchers should include any source of funding (e.g., departmental, commercial, non-commercial, governmental) The HREC will consider whether there is a conflict of interest *(Refer to National Statement 1.1(f), 2.2.6(h), 5.2.8)*

Has this study received research funding or is this submission being made as part of an application for research funding?			☐ Yes	☒ No
What is the status of the funding application?	☒ NA	☐ Approved	☐ Refused	☐ Pending

4.2 The research is to be funded by:
NIL

☐ Government grant or funds
☐ Not-for profit grant or funds
☐ JCU grant or funds
☐ Private sector grant or funds
☐ Student grant
☐ Student in-kind funds, e.g. student's computer, space
☒ JCU in-kind funds (e.g. JCU computer, space, equipment) Please list: JCU computers, laboratory, SPSS license
☐ Other – Please explain:

4.3 What is the approximate amount of funding?

N/A

4.4 If a grant, what is the project grant title, name of fund and proposed funding duration?

N/A

4.5 How will participants be informed of the source of the funding?

N/A

4.6 If the funding is from the private sector, have any conditions been placed on the research, or are there any possible issues pertaining to conflict of interest, e.g. publication restrictions, intellectual property

N/A

5. AIMS AND BACKGROUND

5.1 State the aims and key research questions, any hypotheses to be tested and the significance of the research.
Outline the background and relevance of the project to the field of research, including how the research is justified based on the current literature and its potential to contribute to existing knowledge. Provide references as an attachment to this application to support your answer **(Max 300 words)**
(Refer to National Statement 1.1(c))

In two seminal research articles, Wegner and colleagues (Wegner & Erber, 1992; Wegner et al., 1987) demonstrated that individuals suppressing a target thought ironically experience higher levels of preoccupation with that thought compared to those deliberately concentrating on the same thought. Two types of 'ironic effects' were observed: greater preoccupation of the target thought during and after active thought suppression, known as the immediate enhancement and rebound effects, respectively. A meta-analysis of thought suppression studies revealed significant sizes for the immediate enhancement ($d = 0.34$) and rebound ($d = 0.19$) effects, respectively (Wang et al., 2020). However, none of the identified studies directly replicated the immediate enhancement effect, and only three provided under-powered conceptual replications yielding inconclusive results. In addition, all but two studies testing the rebound effect were under-powered, and the adequately-powered studies yielded effects opposite to the hypothesized direction. There is also evidence that the meta-analytic estimate of the rebound effect may be an artifact of publication bias (Wang et al., 2020). These findings cast doubt on the size and variability of ironic effects identified by Wegner et al. The present research therefore seeks to conduct direct replications of the two seminal studies, along with other laboratories in the world.

6. PARTICIPANT DETAILS

6.1 Describe the target group(s) who will be recruited into this research. *(Refer to National Statement Section 4 and 1.4(a))*

JCUS undergraduate psychology students

6.2 Outline how many participants will be recruited, their sex and age range, experimental/control groups, sampling strategy and any inclusion/exclusion criteria that will be applied. What is the basis for these numbers? *e.g. statistical test, etc.* Please explain the reason behind any proposed or expected imbalances in sex or age and outline what measures will be put in place to prevent bias, or explain how bias is not of concern to the design of the study in the results. *(Refer to National Statement 3.1.14-15)*

We hope to recruit approx. 200 participants in total. This is based on power analyses outlined in our provisionally accepted pre-data manuscript at *Advances in Methods and Practices in Psychological Science* (AMPPS; available at our OSF page: https://osf.io/u3nza/) Gender and age will not be controlled, as this is a replication of the seminal studies in which these variables were not controlled. All participants will be randomly allocated to the experimental conditions.	
How many participants do you need to meet your project's aims?	200
How many participants will be JCU students?	200
Will any participants be Aboriginal or Torres Strait Island peoples? If so, how many of the total sample are expected to identify as Aboriginal and Torres Strait Islander?	☒ No

6.3 Does the research involve participants under the age of 18 years old? If 'YES', please provide details below and attach the document	☐ Yes	☒ No

to this application, of an Australian working with children approval (Blue Card if working in Queensland (JCU Australia), or a Singapore Police Certificate of Clearance (JCU Singapore) or other working with children approval if working in another country *(Refer to National Statement Ch.4.2)*		

7. METHODS

7.1 Provide a brief outline of each step of the proposed methodology, including research design, details of data collection techniques, tasks participants will be asked to complete, the estimated time commitment involved, and how data will be analysed. If the project includes any procedure that is already established and uses accepted techniques please include a description of the procedure. If the study has multiple phases or continues over a considerable time period please consider submitting the phases in separate applications (Max 1 page)
(Refer to National Statement 1.1(b), 1.3, Section 3)

Experiment 1

Participants and Design

100 participants will be randomly allocated to a 2 (suppression manipulation: suppress vs. concentrate) X 4 (target word: mountain vs. car vs. child vs. house) between-participants design. Note that given the complexity of the analytical design for both experiments, the planned analyses that have been accepted at AMPPS are very detailed, so please see Appendix A wherein we have appended the relevant sections from the accepted pre-data manuscript.

Procedure and Stimuli

Participants will participate individually in a computerized lab room equipped with a video recording device (e.g., mobile phone) that captured the computer screen. The experimenter will first obtained informed consent, and then deliver the instructions of the first task – participants in the suppression condition will be asked not to think about a target for the next five minutes. One of the following target thoughts will be randomly provided to each participant: *mountain, car, child,* or *house*. Participants in the concentration condition will be asked to think about one of these targets for the next five minutes. The experimenter will then leave the room for five minutes, and upon returning, will instructed the participants to engage in a concurrent task while continuing the suppression/concentration task. The experimenter will tell the participant that this new task requires them to generate and verbalize clearly an associate word in response to each word presented on the computer screen. Participants will also be told that they should try and provide a response within the time limit, which may be 3 or 10 seconds, as presented beneath the word stimulus. Participants will then complete four practice trials in the presence of the experimenter, following which the experimenter will start the video recording and leave the room for the participant to complete the actual trials. Upon task completion, participants will see a screen that instructs them to knock on the door to signal the experimenter, who will then return and ask them to complete a handout with a probe of their awareness of the aims of the present study followed by another handout with the manipulation check and demographic questions. Finally, participants will be handed the debriefing sheet, and thanked. Note that the materials and informed consent documents for both studies are appended in Appendices A and B respectively, and both studies will take approx. 15 minutes.

Experiment 2

Participants and design

100 participants will be randomly allocated to a 2 (instructions: suppression vs. concentration) X 4 (target thought: gorilla vs. tiger vs. food vs. diseases) between-participants design.

Procedure and Stimuli

Participants will participate individually in a laboratory room equipped with an audio recording device. Participants will first be given instructions on how to verbalize their stream of consciousness. They will be told that the experiment involves several periods during which they should verbalize to the recording device everything that comes to mind. After a practice round, the

experimenter will return, stop the recording, and deliver some instructions based on participants' random allocation. The instructions are:
In the next few minutes, please verbalize your thoughts as you did before, with one exception. This time try (or try not; depending on participant condition) to think of a gorilla/a tiger/food/diseases (depending on participant condition). Every time you mention gorilla/tiger/a food/a disease or have gorilla/tiger/a food/a disease come to mind, please ring the bell on the table before you.
The experimenter will then start recording, say "begin", and leave the room. After five minutes, the experimenter will return, stop the recording, and give a final set of instructions as follows:
Thanks for completing this task, there is just one more round left. So for the next few minutes, please just verbalize your thoughts as you did before, that is (or but this time; depending on participant condition), try to think of a gorilla/a tiger/food/diseases (depending on participant condition). Every time you mention gorilla/tiger/a food/a disease or have gorilla/tiger/a food/a disease come to mind, please ring the bell on the table before you.
The experimenter will then start recording, say "begin", and leave the room. After five minutes, the experimenter will return, stop the recording, and ask participants to complete a handout. This handout contains an additional measure ("looking back on the last two rounds, what percentage of time do you think you spent thinking about a gorilla/a tiger/food/diseases (depending on participant condition)? Please answer with a percentage between 0 and 100 for each of the last 2 rounds"). Finally, participants will be asked to complete the suspicion probe, manipulation check, and demographic questions as per Exp1. They will then be handed the debriefing sheet and thanked.

7.2 Video/Photography

If you are recording either a moving image or photographing your participants, please explain why this is necessary in relation to the research aims of the project. If you are recording images which identify individuals please explain how this action adds to your data collection and validation of findings and ask participants also to sign the JCU Marketing Release Form.

In Exp 1, the computer screen will be recorded so that the results could be coded. We will not be recording images that identify participants.

8. RESEARCH SITES/LOCATIONS

8.1 Outline the research locations and or sites/locations where the research will be conducted and how they will be appropriate to support the needs of the research, physical and emotional needs and safety of the participants and researchers. *(Refer to National Statement 1.1(f))*

As with most laboratory studies conducted at JCUS, this study will be conducted at JCUS in the psychology laboratories.

Overseas Research

8.2 If the data is to be collected from a country outside of Australia (JCU-A) or Singapore (JCU-S), which country or countries will be involved? *(Refer to National Statement Ch. 4.8)*

N/A
Have you received government approval to conduct research in that country? If not, please explain below
• Provide with your application any foreign ethics (or equivalent approvals) and evidence of the foreign government approval for your project to take place in that country • If no ethics approval needed in that country then attach evidence of this, e.g. an email from a collaborating institution, reference from the country's government website, etc

9. RECRUITMENT

9.1 Describe the process for recruiting participants in detail, including how they will be identified and selected, invited to participate and how you will access their contact details, details of any

external organisations assisting, third parties, social media used (include details of groups within social media sites targeted), websites, public documents, details of any personal contacts. *(Refer to National Statement Ch. 1.4(b), 3.1 Element 2)*

The study will be advertised on SONA and JCUS students looking to accrue SONA research credits will be able to sign up and participate at their preferred time slot.

10. CONSENT

10.1 Describe how consent will be obtained from participants (written, verbal, participant & guardian, etc). *(Refer to National Statement Ch 3.1 Element 3)*

An information sheet will be provided to participants, following which they will decide whether or not they would like to sign the consent form and proceed with the study.

Type of consent (e.g. active, specific, extended, unspecified)

10.2 Does the research target participants from any of the groups outlined in *Section C* of the *Risk Assessment Checklist?*	☐ Yes	☒ No
If 'YES', outline for each of those groups you have identified how you ensure the participants provide informed consent, for example: • Provide translations of information sheets and consent forms • Use an interpreter • Seek parental/guardian consent for children • Ensure your research is undertaken in a culturally sensitive manner • Ensure participants understand their participation is voluntary		

10.3 Will the research be requiring a waiver of consent or use an opt-out consent process? If yes, justify this choice based on the criteria outlined in the *National Statement Ch 2.3 – 2.3.2 (limited disclosure) 2.3.10 (waiver) or 2.3.6 (opt-out)*	☐ Yes	☒ No

10.4 Reimbursement/Incentives

Note that while participants may, in certain circumstances, be paid or reimbursed for their inconvenience and time, any payments or inducements that are offered must not be disproportionate to the time involved such that it might encourage participants to take risks. *(Refer to National Statement 2.2.10-11)*

Will the participants be offered any reimbursement of costs, incentives or other rewards proposed to be given to participants?	☒ Yes	☐ No

If ***yes***, describe how much and in what form the payment/incentive will take.

As per the information sheet, participants will receive 2 research credit points

10.5 Dependent/Unequal Relationships and Coercion

The consent of a person to participate in research must not be subject to any coercion. Research involving those in dependent or unequal relationships (e.g. teacher/student, manager/employee, parent/child, doctor/patient) may compromise a participant's ability to give consent that is free from any form of pressure (real or implied) arising from this unequal power relationship.

Does a dependent or unequal relationship exist between any participant and researcher, particularly those involved in recruiting?	☐ Yes	☒ No

If ***yes***, please explain the relationship and the steps to be taken by the researchers to ensure that the participant's participation is purely voluntary and not influenced by the relationship in any way. *(Refer to National Statement Ch. 4.3)*

10.6 English as a Second Language or with Low Literacy Level

Describe how you will ensure that participants, whose first language is not English or who have low literacy level will be able to provide informed consent. *(Refer to National Statement 5.2.17(c))*

University students at JCUS are assumed to have English proficiency. The present study and information sheet also does not involve any significant amount of jargon or technical terms.

10.7 Discontinuing Participation/Withdrawing Consent *(Refer to National Statement 2.2.6(g), 2.2.8, 2.2.19-20)*

Describe how, as part of the informed consent process, participants will be advised that they have the right to withdraw at any time or withdraw any unprocessed data previously supplied.
Note the implications for anonymous surveys where potential participants need to be aware that if they withdraw, their data cannot be identified to be deleted.

The information sheet will specify that: "Taking part in this study is completely voluntary and you can stop taking part in the study at any time without explanation or prejudice".

11. RISK MANAGEMENT AND BENEFITS

11.1 Based on the answers in the Risk Assessment Checklist, outline how any of the identified risks will be addressed according to the *National Statement Chapter 2.1*.

Potential risk	Risk minimisation/management strategy
None	There is no predicted risk because the study simply involves several non-sensitive tasks and questions.

11.2 Potential Benefits

Outline the potential benefits of the research for the following: *(Refer to National Statement 1.1(a), 1.6, 1.8, 1.9)*

The participants
Given that ironic phenomena of thought suppression occupy prominent places in the world of psychology, these concepts are naturally also taught frequently in undergraduate psychology courses. Conducting a multilab replication of ironic effects would therefore benefit researchers, educators, and students, as it would supplement the existing findings by providing an updated, large-scale, cross-cultural, and more methodologically rigorous test of the same phenomena.

The community
As above

11.3 Risk-Benefit Statement

Please give your assessment of how the potential benefits to the participants or contributions to the general body of knowledge would justify any risks. *Even if the risk is negligible, the research must bring some benefit to be ethical. (Refer to National Statement 1.1(a), 1.6-1.9)*

The benefit has been outlined above, which, in contrast to the zero risks presented, suggest that the research is worthwhile.

12. DATA AND INFORMATION MANAGEMENT

Confidentiality of Data and Participant Personal Information

12.1 Human research data must be managed using the Research Data JCU platform and according to JCU's Research Data Management policies and procedures.

Read the *National Statement Section 3 Element 4*, refer to the Research Data Management Toolkit and please ensure that you have read the *Code for the Responsible Conduct of Research* and *Management of Data and Information in Research.*

Indicate how the data, materials and records will be kept to protect the confidentiality, privacy or identities of participants and their data, including hard copies, electronic files and forms.
(Please note the difference between data being *anonymous (no identifying information collected)* and *confidential (identifying information collected, but kept secret)* in your answer.)

☐	Data will be entirely anonymous
☒	Data and records will be coded and non-identifiable
☐	Data and records will be coded and re-identifiable
☐	Some or all of the retained data and records will include personally identifying information
☐	Other, please describe:

12.2 In what form(s) will data be collected and stored? e.g. electronic, hard copy, recordings, etc

Data will be collected via hard copy handouts but coded. The hard copies will be stored in a safe cabinet at JCUS only accessible by the PI, and the electronic data will be stored electronically on Open Science Framework.

12.3 Where will data, materials and records be stored during and after completion of the project and how will it be secured? Provide full details of the location for all types of data.
Refer to JCU Research Data's advice on data storage and compliant data storage options.

The hard copies will be stored in a safe cabinet at JCUS only accessible by the PI. For electronic data, I have registered this project in the JCU RDMP, and data will be stored accordingly

12.4 Who will be responsible for the security of, and access to confidential data and records, including consent forms, collected in the course of the research?

The PI

12.5 The *Australian Code for the Responsible Conduct of Research*, *Section 2.1.1* and JCU Library give the following guidance:

- In general, the minimum recommended period for retention of research data is 5 years from the date of publication.
- Research data that results in a patent must be retained for a minimum of 7 years
- For most clinical trials, data should be retained for a minimum of 15 years, and longer if necessary.
- If the research work has community or heritage value, research data should be kept permanently, preferably within a national collection.

Indicate the minimum period for which data will be retained.

☐	Indefinitely
☒	5 years post-publication
☐	7 years post-publication

☐	15 years post-publication
☐	Other, please describe:

12.6 Is there any reason NOT to share the de-identified or anonymous data from this project? (If there is a chance the data will be shared and re-used for projects in the future, please ensure that there are the appropriate statements on the Participant Information Sheet and Consent Form)

At this stage, we don't foresee any reason that the data collected from this project would not be able to be shared / reused for future projects. Participants will be informed that their data may be used in future projects.

13. PRIVACY

13.1 Please give attention to privacy implications and compliance with legislative requirements including the Privacy Act (Australia), Information Privacy Act (Queensland) or the Personal Data Protection Act (Singapore). Please see the *Guidelines Approved under Section 95A of the Privacy Act 1988* produced by NHMRC. See Queensland Health for information on applying for access to information held by Queensland Health and submitting a PHA application.

a) Does this project involve gaining access to medical information from a Commonwealth agency? If 'Yes', which agency?	☐ Yes	☒ No
b) Does this information require the disclosure of personal information, i.e. identifiable information? If 'Yes', what type of information will you be accessing?	☐ Yes	☒ No
c) If you answered YES to the previous question – Will you obtain informed consent from the individuals to whom the information is related? If you are not obtaining consent, please explain why not?	☐ Yes	☐ No

d) Does this project involve the collection, use or disclosure of health information from a private sector organisation? If 'Yes', which private sector organisation?	☐ Yes	☒ No

13.2 Is the data from the private sector organisation going to be used for research which is related to:

• Research relevant to public health or safety?	☐ Yes	☐ No
• The compilation or analysis of statistics relevant to public health or safety?	☐ Yes	☐ No
• Management, funding or monitoring a health service?	☐ Yes	☐ No
• Will you obtain informed consent from the individuals to whom the health information is related? If 'No' explain why not?	☐ Yes	☐ No

14. DISSEMINATION OF RESULTS

14.1 Explain when, how, where and to whom results will be disseminated, including whether participants will be provided with information on the project's findings or outcomes. *(Refer to National Statement Ch. 3, Elements 6 and 7)*

Findings will be published in a peer review journal which should be accessible by participants and anyone affiliated with a tertiary educational institution

14.2 How will results be made available to participants?

☐ Written summary of results
☐ Copy of final manuscript (thesis, paper etc)
☐ Verbal presentation (info session, debriefing etc)
☐ Presented to all participants
☒ Presented if requested
☐ Presented to representative of participants (CEO, teachers of students)
☐ Other – Please explain:
☐ The results will not be shared with participants – Please explain:

15. Supporting Documentation Checklist

If the recruitment methods/participants involve any of the following please provide a copy of the corresponding documents with your application:

☐	Mail-out	☐	Copy of the document to be mailed
☐	Email	☐	Copy of the email wording
☐	Telephone	☐	Script of the telephone conversation
☐	Advertisement	☐	Copy of the advertisement
☐	Social media	☐	Permission from social media and wording of post
☐	Recruiting through third parties	☐	Letter of support from third party, including details of their participation

16. DECLARATIONS

16.1 Principal Investigator

I declare that (indicate by ticking boxes):

☒ All investigators of this research project are qualified and authorised to perform procedures described in this document

☒ The assistants involved in this project have been fully briefed on procedures and relevant ethical considerations

☒ I am aware of the responsibilities set out in the relevant legislation

☒ I undertake to inform the Human Research Ethics Committee (HREC) of any changes to the proposed procedures or details given in this form subsequent to its submission (including change of contact details)

☒ I agree to assist the Committee to monitor the conduct of research by completing and promptly returning annual reports and provide a final report upon completion of the project as appropriate

☒ This project complies with the National Health and Medical Research Council *National Statement on Ethical Conduct in Human Research, 2007 (Updated 2018)*

☒ The purpose of this project cannot be achieved by alternatives to the use of human participants

Research Integrity Training

Please ensure that you complete the mandatory Research Integrity Training for Researchers in LearnJCU (if you have trouble accessing this training, please contact researchintegrity@jcu.edu.au)

	Name	**Signature**	**Date**
Principal Investigator	Deming (Adam) Wang		25/11/2022

Student Supervisor			
Student Supervisor			

16.2 Dean

I declare that:

- There are adequate resources, facilities and support for this project to proceed to conclusion
- The staff members involved are appropriately qualified and competent for the task described
- Adequate instructions have been given for participant welfare and post-project care and monitoring

Position	Name	Signature	Date

Human Research Ethics Advisor's Review

Instructions for Researchers

1. Please send the unsigned, word version of your HREC Application to a Human Research Ethics Advisor for review before the meeting.
2. After you have received their review, make any changes to the application suggested by the Advisor, please highlight these changes in yellow so the HREC can see where the feedback has been addressed.
3. If you do not agree with the feedback, please provide a written response outlining the reasons with your application.
4. Then obtain the appropriate signatures before submitting the final version and a copy of this review as one PDF document to ethics@jcu.edu.au
5. Please don't send it back to the Ethics Advisor for re-review unless they have specifically asked you to.

Instructions for Human Ethics Advisors

1. Please review the application and provide any advice to the researcher that you feel will correct any ethical concerns, improve the application and make it align with the principles of the *National Statement* and other applicable guidelines or legislation.
2. Outline your comments, advice and suggestions to the researchers below, using the following headings and the sections of the *National Statement* cited in the application form as a guide, and return to the researcher as a PDF. If the information in the section is acceptable and no changes need to be made, make a comment to that effect.
3. You are only required to review each application once; you don't need to review revised versions of the same application.

Title					
Appropriate					
General Comments					
Appropriate					
Research Team and Experience					
Clarification as to whether there are collaborators, as this is a multi-lab study.					
Facilities/Settings					
Appropriate					
Participants					
Appropriate – note to add the SONA credits for participant incentives					
Aims, Background and Methods					
Appropriate – note to add estimated time taken for the experiments in section 7.1					
Recruitment and Consent					
Appropriate					
Risk Management and Benefits					
Appropriate					
Data and Information Management and Privacy					
Appropriate					
Other					
Requested amendments to participant information sheet and consent form, informing them that there would be video/audio recording of their computer screen and voices.					
Advisor's Name	**Sherry Aw**	**Signature**		**Date**	**6th Dec 2022**

Principal Investigator response to Ethics Advisor review

Title
NA
General Comments
NA
Research Team and Experience
Note that collaborating labs would need to apply for ethics approval through their own institution and hence were not included in the present application
Aims and Background and Methods
This is already in the information sheet, but I have reiterated it in this application
Facilities/Settings
NA
Participants
This is already in the information sheet, but I have reiterated it in this application
Recruitment and Consent
NA
Risk Management and Benefits
NA
Data and Information Management and Privacy
NA
Other
The suggested content has now been added to the information sheet and consent form

References

Peirce, J. W., Gray, J. R., Simpson, S., MacAskill, M. R., Höchenberger, R., Sogo, H., Kastman, E., Lindeløv, J. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods.* https://doi.org/10.3758/s13428-018-01193-y

Wang, D., Hagger, M. S., & Chatzisarantis, N. L. D. (2020). Ironic effects of thought suppression – A meta-analysis. *Perspectives on Psychological Science, 15,* 778-793. https://doi.org/10.1177/1745691619898795

Wegner, D. M., & Erber, R. (1992). The hyperaccessibility of suppressed thoughts. *Journal of Personality and Social Psychology, 63,* 903-912. https://doi.org/10.1037/0022-3514.63.6.903

Wegner, D. M., Schneider, D. J., Carter, S., & White, T. (1987). Paradoxical effects of thought suppression. *Journal of Personality and Social Psychology, 53,* 5-13. https://doi.org/10.1037//0022-3514.53.1.5

Appendix A

Exp 1 computer task stimuli details:

A total of 16 words from three categories will be presented in random order (programmed using PsychoPy; Pierce et al., 2019) via the Pavlovia online platform (https://run.pavlovia.org/vespr/wegner-replication). Four of the 16 words will be semantic associates of the target word (e.g., *mother, little*, *adult,* and *young* for the target word *child*). Another four words will be nontarget words, which consist of the four semantic associates of a different target word (e.g., *home, door, brick,* and *roof* if the target word was *child*). The remaining eight words will be fillers unrelated to any of the four target words (*shallow, shave, gesture, idea, quarter, anger, voice,* and *option*). In line with the original study, two versions of the computer task were programmed, each with a different pair of target words. Specifically, for participants allocated the target word of either *mountain* or *car*, the first version of the online program will be used. This way, the associates of *mountain* (*climb, high, hill*, and *top*) will be the nontarget words for *car,* while the associates of *car* (*bus, drive, truck,* and *wheel*) will be the nontarget words for *mountain*. For participants allocated the target word of either *child* or *house*, the second version of the online program will be used. This way, the associates of *child* (*mother, little*, *adult,* and *young*) will be the nontarget words for *house,* while the associates of *house* (*home, door, brick,* and *roof*) will be the nontarget words for *child*.

Exp 1 analysis details from accepted pre-data manuscript:

Results

XXX labs (total N = xxx) participated in Study 1. We analyzed the data for each lab (data from Studies 1 and 2 were analyzed independently), before combining all data in a meta-analysis. The analysis code for the analyses is available on https://osf.io/u3nza/. Data

from all labs completing the study consistent with the preregistration (including minimum required sample size post-exclusion) were included in our analysis. The exclusion criteria included failing suspicion checks (k = xxx) and non-compliance (k = xxx).

Main analyses

The dependent variable was the number of prompted responses across trials, and this was dependent on the trial type. For target-prompted trials, a prompted response was counted when participants mentioned the word that they were supposed to suppress or concentrate on. For example, for participants who were supposed to suppress or concentrate on *child*, any mentions of *child* in response to *mother, little*, *adult,* or *young* were counted. For the nontarget-prompted trials, which served as a baseline, for participants who were supposed to suppress or concentrate on *child*, any mentions of *house* in response to *home*, *door*, *brick*, or *roof* were counted.

We first report results using a fixed effects analysis of variance (ANOVA) so that we can better mirror the analyses made by the original authors. In this spirit, a preliminary 2 (instructions: suppress vs. concentrate; between-participants) x 4 (target word: mountain, car, child, house; between-participants) x 2 (word type: target vs. non-target; within-participants) x 2 (time pressure: 3s vs. 10s; within-participants) ANOVA was conducted, the goals of which were twofold. First, this analysis confirmed an absence of interaction effects of target word. Second and expectedly, we found no reliable effects of instructions or time pressure on non-target prompted responses. As a result, in further analyses, we collapsed the results across the four target words, and also focused only on the results of the target word type conditions, essentially simplifying the main analyses down to a 2 (instructions: suppress vs. concentrate; between-participants) x 2 (time pressure: 3s vs. 10s; within-participants) design. Note that we proceeded with the simplified analyses as outlined above regardless of the results of the preliminary analyses.

Exp 1 and Exp 2 suspicion probe:

Before we move on, if you think you know the study's aim already (e.g., if you have participated in a similar study before), please write below what you think the aim is. If you are unaware or unsure, please simply write "X".

_____

_____

_____

Exp 1 manipulation check:

Please answer the question below by circling your answer:

I was asked to _____ about the word that I was given.

think not think

Exp 1 and Exp 2 demographics questions:

What is your age? _____

What is your gender? _____

Exp 1 feedback sheet:

The study that you just took part in actually investigated whether compared to concentrating on a thought, suppressing the same thought would make people more sensitive to that thought under time pressure. We randomly allocated you to either a concentration or suppression condition by telling you to either think about or not think about a particular thought. Using the computer task, we examined whether suppressors were more likely to respond with the target thought compared to concentrators, especially under high time pressure. We hope that our findings can shed some more light on how our mental control efforts can sometimes ironically have counterproductive effects. Thank you for your participation. Please do not

hesitate to contact Dr Adam Wang (adam.wang@jcu.edu.au) if you have any questions or would like a summary of the results of this study.

Exp 2 instructions for stream of consciousness task:

This study is concerned with what and how people think. During several periods you will be asked to describe your stream of consciousness, to indicate what is going through your mind. The following ground rules apply to every aspect of the study. (1) Measures have been taken to ensure your privacy and to guarantee confidentiality concerning your participation in this study. To be specific: (a) you were assigned a number which will be the only identifying mark on all of the data-gathering items in this study; (b) there is no "key" or master-list linking your name to your subject number or to any of the data-gathering materials. (2) When asked to report on your thoughts, please convey whatever information you can on your stream of consciousness at that moment. Your report might include (but is not limited to) descriptions of: images, ideas, memories, feelings, fantasies, plans, sensations, observations, daydreams, objects which catch your attention, efforts to solve a problem. There are no restrictions, qualifications, conventions, or expectations: simply report on whatever is going through your mind (whatever you are conscious of or aware of). If you have any questions, please ask.

Exp 2 manipulation check:
Please answer the question below by circling your answer:

In the second last phase, I was asked to _____ about a particular thought.

think not think

Exp 2 analysis details from accepted pre-data manuscript:

Results

XXX labs (total N = xxx) participated in Study 2. We analyzed the data for each lab (data from Studies 1 and 2 were analyzed independently) before combining all data in a meta-analysis. The analysis code for the analyses is available on https://osf.io/u3nza/. Data from all labs completing the study consistent with the preregistration (including minimum required sample size post-exclusion) were included in our analysis. The exclusion criteria included failing suspicion checks (k = xxx) and non-compliance (k = xxx).

Main analyses

The focal dependent variable was the number of occurrences of the target thought during the second period of the study in the forms of a) target thought mentions only, b) bell rings only, c) both at the same time, and hence d) the combined total of these three measures for each participant.

A preliminary 2 (initial instructions: suppression vs. concentration) X 4 (target thought: gorilla vs. tiger vs. food vs. diseases) ANOVA was conducted. This confirmed an absence of interaction effects of target thought. We therefore collapsed the results across the two target thoughts in the main analyses. Note that we proceeded with this aggregated analysis regardless of the results of the preliminary analyses.

<u>Exp 2 feedback sheet:</u>

The study that you just took part in actually investigated whether compared to concentrating on a thought, suppressing the same thought would make people more preoccupied with that thought. We randomly allocated you to either a concentration or suppression condition by telling you to either think about or not think about a particular thought in an initial period. In a subsequent period, we instructed you to think about the same thought, regardless of your instructions in the initial period. We tested whether people who had initially suppressed a

thought would verbalise it and/or think about the thought more than those who had initially concentrated on the thought. We hope that our findings can shed some more light on how our mental control efforts can sometimes ironically have counterproductive aftereffects. Thank you for your participation. Please do not hesitate to contact Dr Adam Wang (adam.wang@jcu.edu.au) if you have any questions or would like a summary of the results of this study.

Appendix B

INFORMATION SHEET

PROJECT TITLE: a study on thought processes

You are invited to take part in a research project about individuals' thought processes. The study is being conducted by Dr Adam Wang at James Cook University – Singapore campus.

If you agree to be involved in the study, you will be invited to a laboratory task on campus that will take approx. 15 minutes, after which you will receive 2 research credit points. During this experiment, you will be asked to do some thought-related tasks, and answer some quick questions. Note that during the experiment, the computer screen may be video recorded and your voice may be audio recorded.

Taking part in this study is completely voluntary and you can stop taking part in the study at any time without explanation or prejudice, and you can withdraw any unprocessed data provided.

During this experiment, THERE SHOULD NOT BE ANY SIGNIFICANT RISKS OR DISTRESS FOR PARTICIPANTS.

Your responses will be strictly confidential. Once collected, your responses will immediately be tabulated electronically using only your participant number while your consent form (containing your name) will be stored separately from other responses/questionnaires. Both the consent form and other responses will be stored at JCU – Singapore in a locked cabinet and will only be accessible by the primary researcher. The data from the study will be used in research publications and reports, and may be used for future research. You will not be identified in any way in these publications.

If you have any questions about the study or would like to receive a summary of the result, please contact Dr Adam Wang

Principal Investigator:
Dr Adam Wang
James Cook University
Phone: 6709 3746
Email: adam.wang@jcu.edu.au

If you have any concerns regarding the ethical conduct of the study, please contact:
Human Ethics, Research Office
James Cook University, Townsville, Qld, 4811
Phone: (07) 4781 5011 (ethics@jcu.edu.au)

INFORMED CONSENT FORM

PRINCIPAL INVESTIGATOR	Dr Adam Wang
PROJECT TITLE:	A study on thought processes
COLLEGE:	College of Healthcare Sciences (James Cook University – Singapore)

I understand the aim of this research study is **"**to investigate individuals' thought processes**".** I consent to participate in this project, the details of which have been explained to me, and I have been provided with a written information sheet to keep.

I understand that my participation will involve a laboratory task on campus (JCU Singapore) that **will take approx. 15 minutes**, during which I will be asked to do some thought-related tasks and answer some quick questions.

I acknowledge that:

- taking part in this study is voluntary and I am aware that I can stop taking part in it at any time without explanation or prejudice and to withdraw any unprocessed data I have provided;

- that any information I give will be kept strictly confidential and that no names will be used to identify me with this study without my approval;

(Please tick to indicate consent)

I consent to doing some thought-related tasks		**Yes**		**No**
I consent to possibly having my computer screen video recorded and my voice audio recorded (face & other identifiable features will not be recorded)		**Yes**		**No**
I consent to answering some quick questions		**Yes**		**No**

Name: *(printed)*	
Signature:	**Date:**